

INDUSTRIAL PRODUCT DATASHEET

Wio-E5 mini Dev Board - STM32WLE5JC, ARM Cortex-M4 and SX126x embedded, supports LoRaWAN on EU868 & US915

Ultra-compact LoRaWAN development board with STM32WLE5JC and SX126x, delivering long-range, low-power connectivity for rapid IoT prototyping on EU868 and US915.

SKU

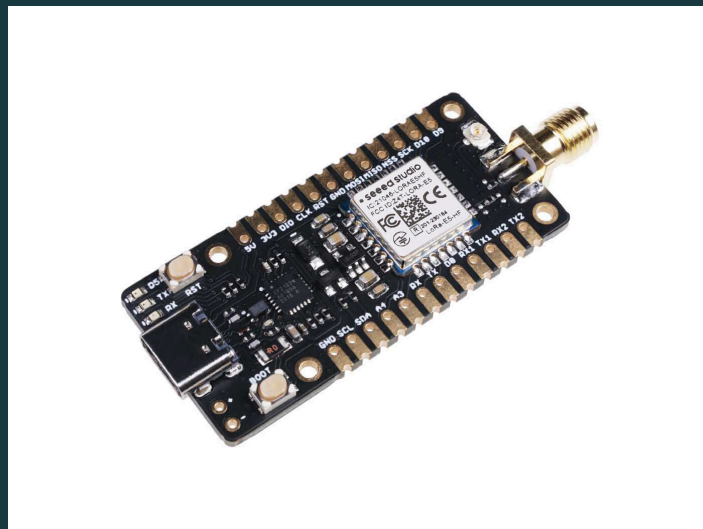
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PRICE

\$21.90(Excl. VAT)



Overview



The Wio-E5 mini Dev Board is a compact, low-power prototyping platform built on the STM32WLE5JC SoC with embedded SX126x, supporting LoRaWAN on EU868 and US915 frequency plans. It features AT command firmware for easy development, fully lead-out GPIOs, RESET and BOOT buttons, and is FCC, CE, IC, and Telec certified, making it suitable for long-range IoT sensor networks.

Key Features

Note

The **Wio-E5 LoRa module** on the board is now IC certified for Canada! New and old products will be shipped randomly.

Wio-E5 mini is a compacted-sized development board suitable for the rapid testing and building of small-size prototyping and helps you design your ideal Long Range wireless IoT device with a long-distance transmission range.

This ultra-compact module has extremely low power consumption and is capable of supporting both (G)FSK and Long Range. With its built-in AT command firmware, makes it easy for developers to build their prototypes or applications. It can easily support LoRaWAN® protocols and a variety of frequency plans, making it an ideal solution for Long Range sensor nodes and a range of wireless communication applications. Whether you need to set up Wireless Meter Reading, IoT Nodes, or a Sensor Network, the Wio-E5 is perfectly suited for low power wide area IoT scenarios.

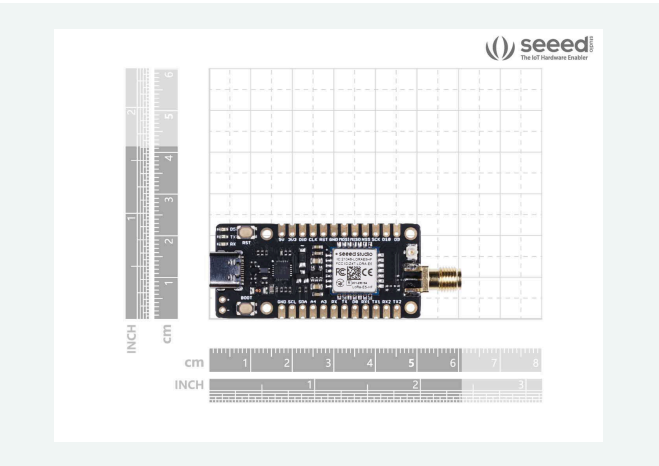
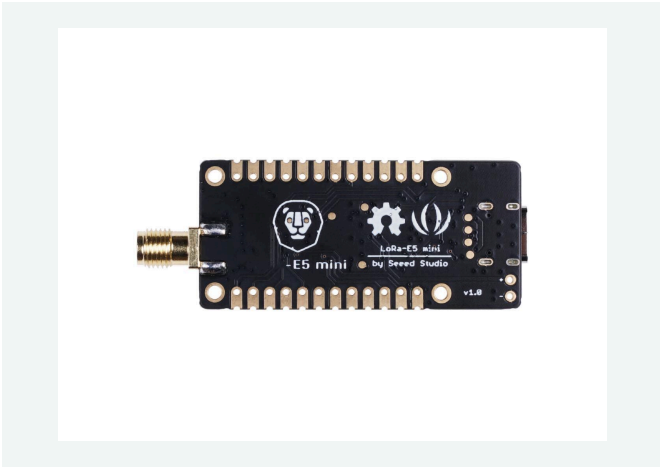
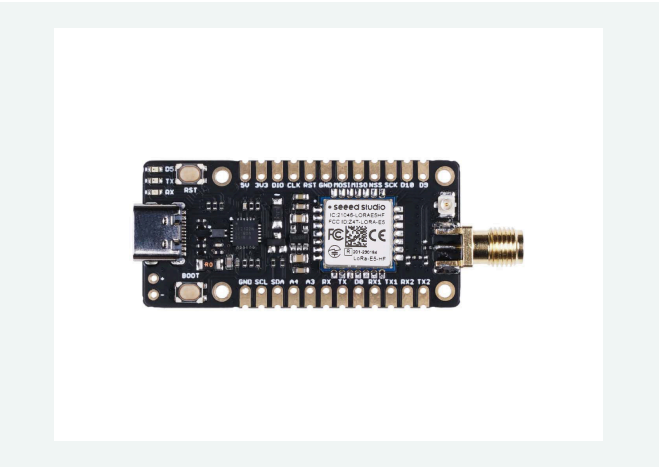
With this module, you can connect to different LoRaWAN server platforms like TheThingsStack, Chirpstack, and others.

Note

If you have a large quantity request or need any further information on Seeed IIoT products and services, please don't hesitate to contact us directly at order@seeed.cc or click [here](#). We'll be happy to offer you our best discount price. We are excited to see you can scale up and expand your business with our products and services.

- **High Performance:** The transmit power up to 22dBm @868/915MHz; -136.5dBm sensitivity for SF12 with 125KHz BW.
- **Long Distance Use:** 158dB link budget
- **Wireless Connectivity:** Embedded Long Range protocol, AT command, support global Long Range frequency plan
- **Use-friendly:** It features fully lead-out GPIOs, as well as convenient RESET and BOOT buttons on board.
- **FCC, CE, IC and Telec Certified**

Product Gallery



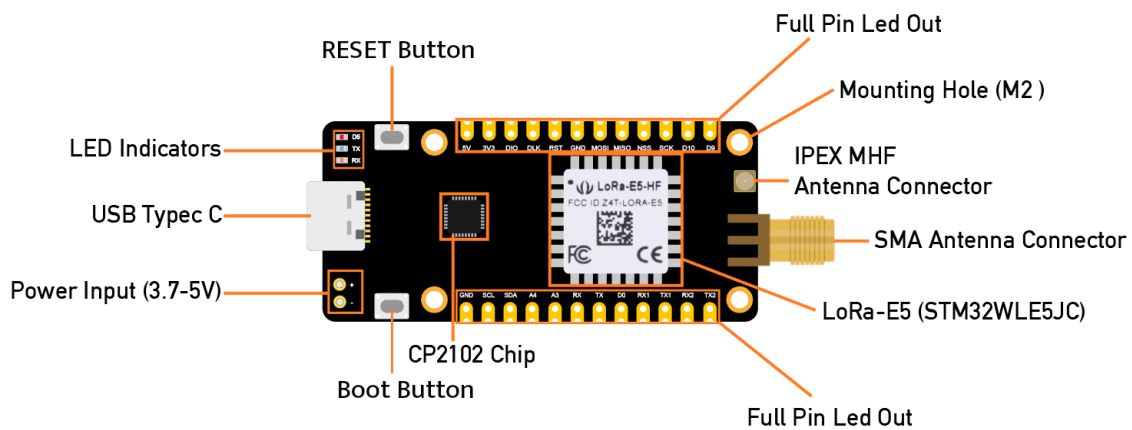
Specification

Parameters	Specifications
size	50*23mm
voltage - supply	3.7V - 5V

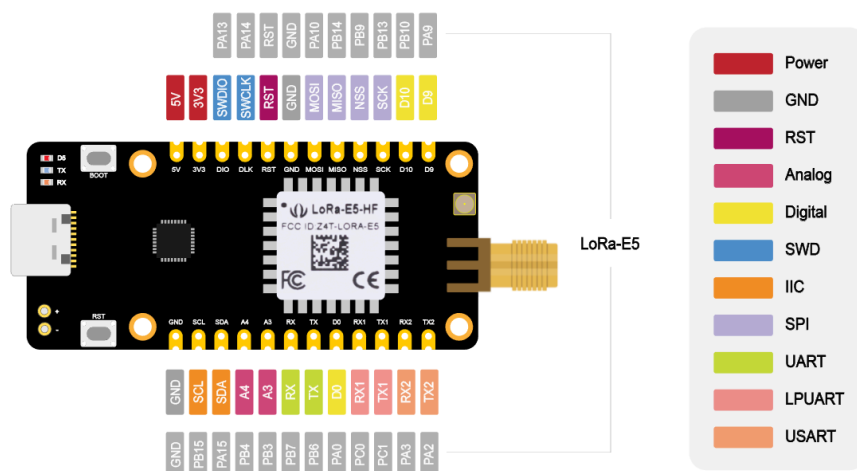
Parameters	Specifications
power - output	up to +20.8 dBm at 3.3V
working frequency	868/915MHz
protocol	Long Range
sensitivity	-116.5 dBm ~ -136 dBm
interfaces	USB Type C / 2P-2.54mm Hole / 1*12P-2.54mm Header*2 / SMA-K / IPEX
modulation	Long Range , (G)FSK, (G)MSK, BPSK
working temperature	-40°C ~ 85°C
current	Wio-E5 module sleep current as low as 2.1uA (WOR mode)

Hardware Overview

LoRa-E5 mini Hardware Specification



LoRa-E5 mini Pinout



Before Packaging Upgrade



After Packaging Upgrade



Note:

The replacement of the old and new packages is being carried out step by step, and there may be mixed delivery of the old and new packages, please kindly understand.

Applications

- **Solar-Powered Outdoor LoRaWAN Air Quality Monitor**
- **Low-Power Snow Depth Sensor Monitoring for IoT Smart City Based On Wio-E5 Applied in Norway**
- **Developed Stream Runoff Monitoring Sensors Using Wio-E5 to Measure The Temperature, Quality and Flow**
- **Smart City: Wio-E5-based Indoor Air Pollutant Detection Project**
- **Loko: The Tiniest GPS Tracker with Built-In Wio-E5 Module Can Track Any Wild Animals and Drones**

Part List

Part List:

Wio-E5 mini *1

Antenna(EU868/US915) *1

USB TypeC (20cm) *1

Sticker *1

1X12pin male pin headers *2

Compliance & Logistics



HSCODE	8517709000
USHSCODE	8517790000
UPC	
EUHSCODE	8517180000
COO	CHINA

Resource Links

- <https://www.seeedstudio.com/LoRa-E5-mini-STM32WLE5JC-p-4869.html>
- [Wio-E5 mini v1.0.brd](#)
- [Wio-E5 mini v1.0.pdf](#)
- [Wio-E5 mini v1.0.sch](#)
- [Wio-E5 datasheet and specifications](#)
- [Wio-E5 AT Command Specification](#)
- [STM32WLE5JC Datasheet](#)
- [Wio-E5-HF Certification CE-VOC-RED](#)
- [Wio-E5-HF FCC Certification -DSS](#)
- [Wio-E5-HF FCC Certification -DTS](#)
- [Wio-E5-HF TELEC Certification](#)
- [Wio-E5-HF IC Certification](#)
- [STM32Cube MCU Package for STM32WL series](#)

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